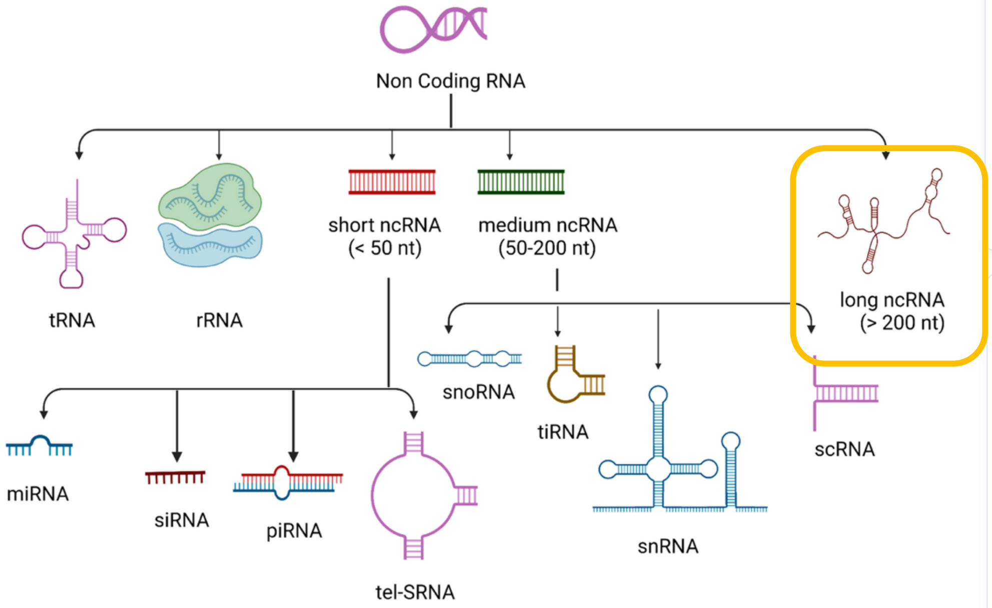
ROTATION PROJECT

lncFit

Predicting lncRNA Essentiality
from gRNA Sequence + Cell Context

Kelly Lee · 6/29/2026

What are lncRNAs?



Long noncoding RNAs (lncRNAs)

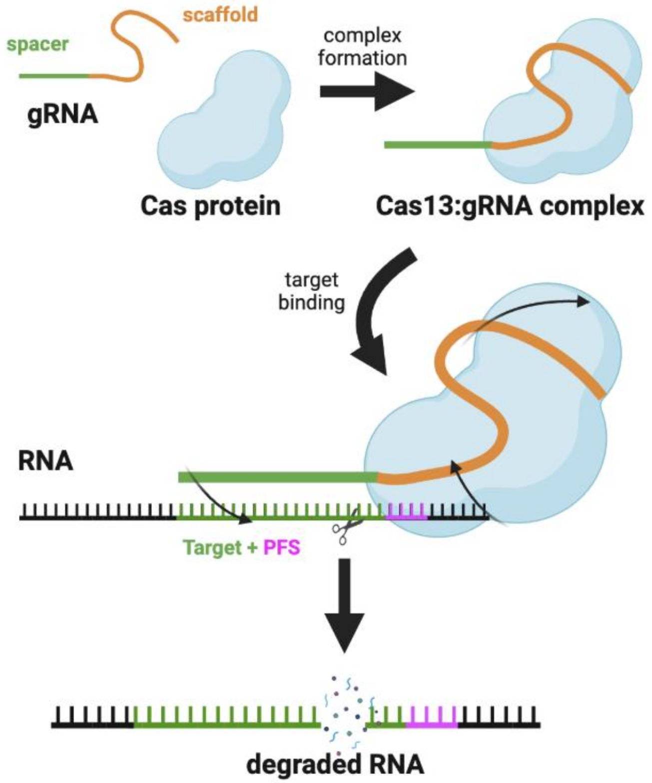
- > 200 nucleotides long, spliced & polyadenylated
- Not translated into protein
- Low sequence conservation across species
- < 1% linked to a clear function — most are uncharacterized

Known roles in cell biology

- 🔄 Cell cycle & proliferation
- 💀 Apoptosis
- 🧬 Chromatin regulation
- 🎯 Cancer & development
- ⚠️ Low abundance, cell-type specific, no protein product

Sharma et al., 2024, Biomed Sci

The dataset: CRISPR-Cas13 screens



Dataset at a glance

5,500	5	788	82%
lncRNAs screened	human cell lines	essential lncRNAs found	cell-type specific

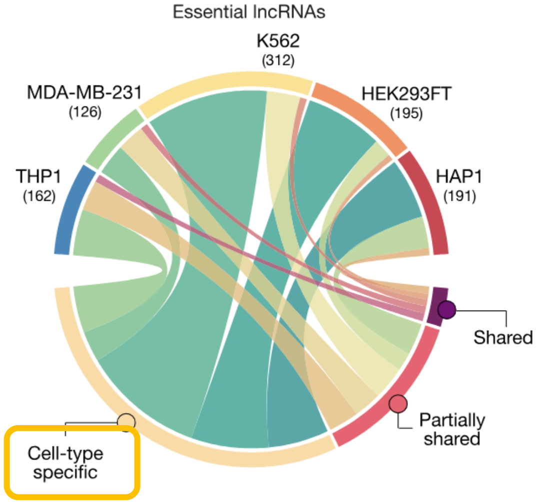
Key biological findings

- 82% of essential lncRNAs are cell-type specific
- Most act independently of their nearest protein-coding gene
- Loss → cell cycle arrest and apoptosis
- Expression correlates with survival in 9,000 primary tumors

Liang et al., 2026, Cell Genomics

Can we predict lncRNA essentiality?

Why the same lncRNA behaves differently



Reasons for cell-type specificity

- Cell-type-specific expression
- Cell-type-specific binding partners
- Different transcriptional programs

The bottleneck

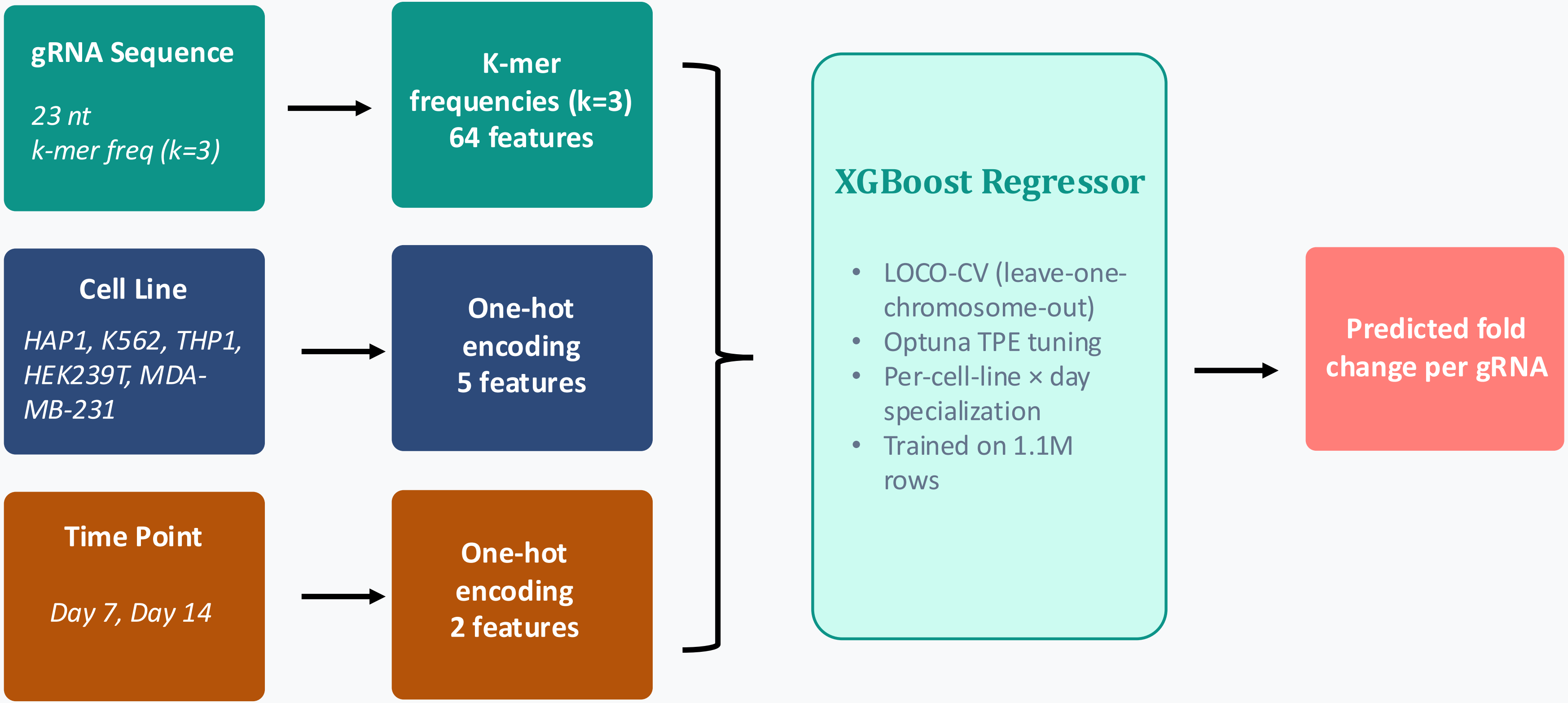
- 788 essential lncRNAs found — in 5 cell lines
- No lncRNA annotation in the ~1,000 other human cell types
- Running Cas13 screens is slow and expensive
- No model to generalize to new contexts

The goal

- **Predict lncRNA essentiality from gRNA sequence**
- **No new screen needed for other cell lines**

Liang et al., 2026, Cell Genomics

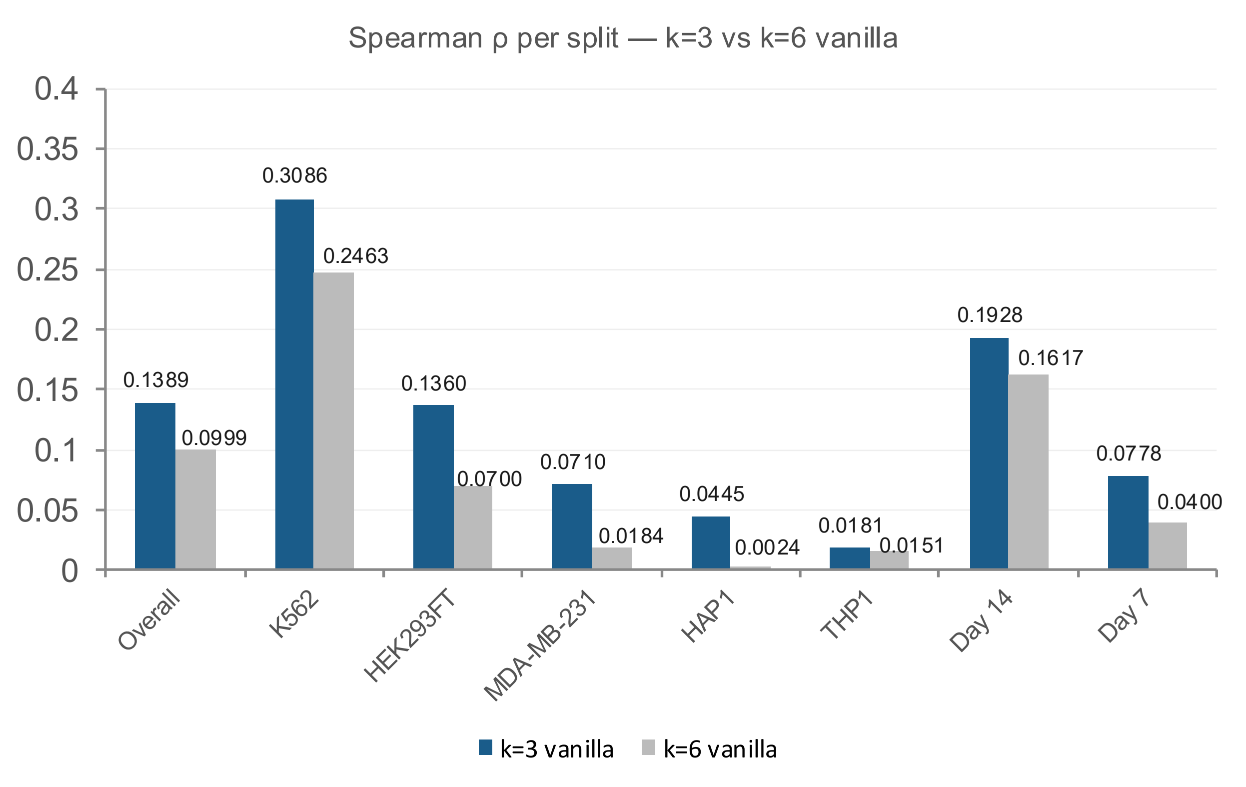
Pipeline overview



Key Findings

k=3 outperforms k=6 on every cell line and both time points

Longer k-mers don't help. K562 drives most signal – THP1/HAP1 near-random in both



K562 is most predictable

- Has the strongest sequence-driven proliferation program
- Is the most studied and best characterized cell line
- Has the most essential lncRNAs

THP1/HAP1 near-random

- Their essentiality may be driven by cell-type-specific biology, not sequence
- Fewer essential lncRNAs

Day 14 is 2.5× Day 7

Depleted cells need time to wash out

Optuna TPE + LOCO-CV baseline

Set up Optuna-TPE hyperparameter tuning with chromosome leave-one-out CV (LOCO-CV); established the first reproducible XGBoost baseline at Spearman $\rho \approx 0.14$ on the held-out chr1 test set (K562 Day 14 reaches $\rho \approx 0.31$).

The protocol

- 23-fold LOCO-CV
- Test set is chr1

The tuning strategy

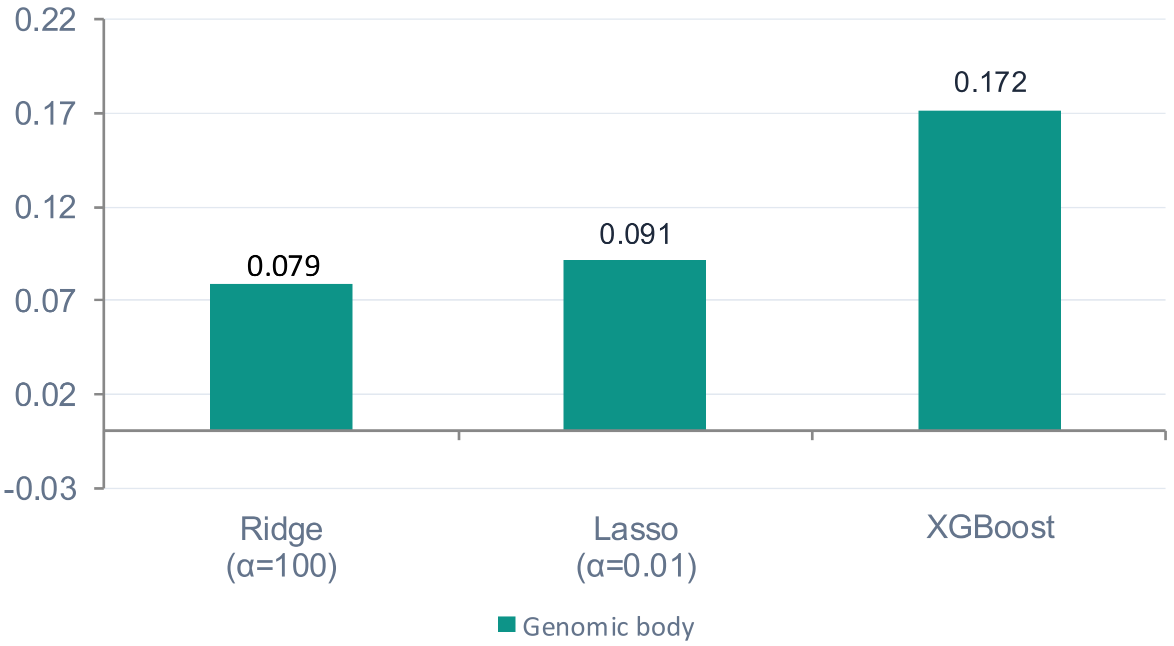
- Optuna TPE with MedianPruner and XGBoost early stopping to auto-pick n_estimators.

The baseline

On the held-out chr1 test set:

- Pearson $r = 0.196$
- Spearman $\rho \approx 0.14$
- $R^2 = 0.019$

XGBoost shows improvement compared to linear baselines



0.172

XGBoost Spearman ρ

vs 0.079 Ridge (~2×)

Genomic ≈ transcript body features — both ~2× linear models. The bottleneck is model class, not feature set.

Per-cell-line training marginally helps 4/5 lines

Cell line	n test	Pooled ρ	Per-CL ρ	Δρ
K562	13,884	0.309	**0.319**	**+0.010**
HEK293FT	13,728	0.156	0.169	**+0.013**
MDA-MB-231	13,896	0.118	0.140	**+0.023**
HAP1	13,888	0.065	0.070	**+0.005**
THP1	13,896	0.042	0.038	**−0.004**

✓ Pooling was marginally diluting signal

4/5 lines marginally improve despite training on 5× less data
MDA-MB-231: largest gain (Δρ = +0.023)
K562 already strongest: ρ = 0.319

✗ Not a pooling problem

HAP1: +0.005 (negligible)
THP1: −0.004 (slightly regresses)
Near-constant target (RMSE ≈ 0.25) suggests noisy labels or sequence-unpredictable biology

Representation bottleneck confirmed — per-cell-line specialization can't fix HAP1/THP1. ChatNT is the key test.

Essentiality depends on cell line and screening day

Cell line	Day 7 ρ	Day 14 ρ	Per-CL best	Best day Δ
K562	0.006	**0.319**	0.319	**+0.000**
HEK293FT	**0.228**	0.084	0.169	**+0.059**
MDA-MB-231	0.110	**0.169**	0.140	**+0.029**
HAP1	0.025	**0.105**	0.070	**+0.035**
THP1	0.026	**0.049**	0.038	**+0.011**

ρ ≥ 0.20 ρ ≥ 0.10 ρ ≥ 0.05 ρ < 0.05

★ Best result in project

K562 Day 14 ρ = 0.319
Strongest single-split signal to date

⚠ Day × cell-line interaction

K562: Day 7 → 0.006, Day 14 → 0.319
HEK293FT flips: Day 7 → 0.228, Day 14 → 0.084

Pooled models miss this

DNABERT-2 body embeddings improve k-mer ceiling

Why add DNABERT-2 sequence embeddings?

The XGBoost baseline throws away:
- positional / motif syntax
- long-range dependencies between distant parts of the transcript

DNABERT-2 is a BERT-style transformer pretrained on multiple genomes at nucleotide resolution.
- Contextual embeddings: motif grammar, codon context, regulatory structure, etc.

Hypothesis: if contextual features carry essentiality signal, appending 768-dim DNABERT-2 vectors should push XGBoost's CV Spearman ρ above the baseline.

Condition	CV ρ (mean ± std)	Δ vs baseline
k-mer only (baseline)	0.133 ± 0.014	—
k-mer + body (first 1000 bp)	0.166 ± 0.008	+0.034
k-mer + body (last 1000 bp)	0.175 ± 0.009	+0.043
k-mer + body (mean)	0.167 ± 0.010	+0.035
k-mer + guide embeddings (23 bp)	0.121 ± 0.013	−0.011
k-mer + body (mean) + guide	0.164 ± 0.009	+0.032

Next Steps

From k-mers to language models

Next: Fine-tune ChatNT for lncRNA essentiality prediction

Why ChatNT?

1. Multimodal by design — takes a natural-language task spec plus a raw DNA sequence in a separate encoder channel
2. Pre-trained on 850 billion DNA tokens — zero-shot sequence understanding
3. But it’s not trained on lncRNA (log2FC regression was not in ChatNT’s original training)

Goal

Adapt ChatNT to predict CRISPR-Cas13 screen log2FC prediction directly from cell line, 23-nt guide pairs

ChatNT prompt

In {cell_line} cells, would targeting the lncRNA represented by these DNA sequence regions <DNA> alter cellular essentiality? Return the predicted CRISPR-screen log2 fold-change (log2FC) as a single numeric value.

Can ChatNT predict lncRNA essentiality (CRISPR-screen log2FC) from DNA sequence, and does QLoRA fine-tuning beat zero-shot ChatNT + an XGBoost baseline on the held-out chromosome-1 test set?

Summary

WE DID

XGBoost + k-mers on 1M+ Cas13 records — best result: K562 Day 14 $\rho = 0.323$

WE FOUND

HAP1 & THP1 near-random even with dedicated models — representation bottleneck, not pooling

SO WHAT

Sequence alone isn't enough — next: fine-tune ChatNT on gRNA prompts

Supplementary

What is a k-mer? What is our feature space?

Sequence: A T G C A T G
k=3 kmers: ATG, TGC, GCA, CAT, ATG

gRNA k-mers: 64 features

- Slide the window across the 23-nt gRNA sequence,count how often each of the 64 trinucleotides appears, then normalize by sequence length

Cell line + day: 7 features

- 5 one-hot encoding for cell lines (K562, HEK293FT, MDA-MB-231, HAP1, THP1)
- 2 one-hot encoding for days (day 7, day 14)

lncRNA body k-mer: 64 features

- First 1000 bp window
- Last 1000 bp window

lncRNA body k-mers

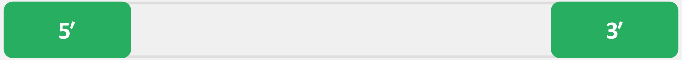
Before — guide sequence k-mers

23 nt guide RNA

Captures: guide efficacy
Misses: lncRNA biology

After — lncRNA body k-mers

First 1000 bp + Last 1000 bp



Captures: promoter-proximal elements, polyadenylation signals, structural motifs

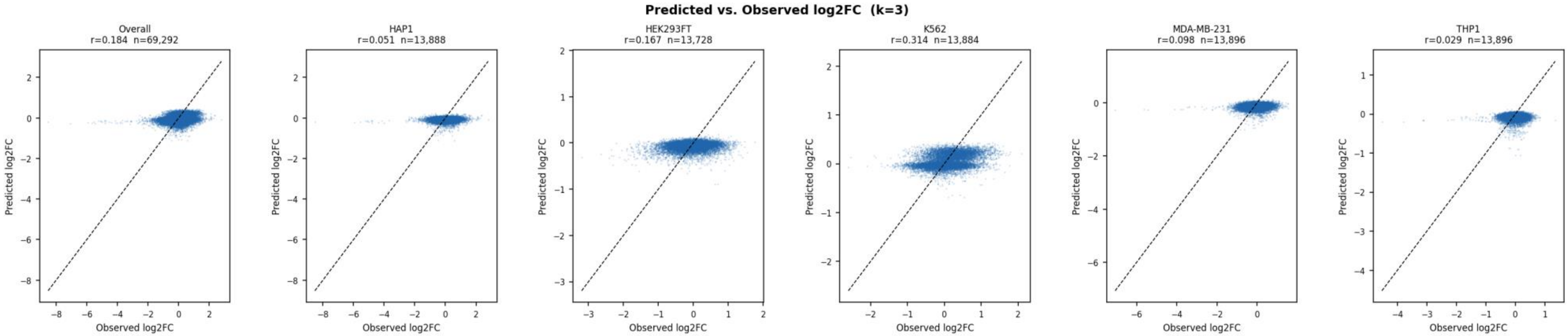
Promoter/TSS-proximal signal

- 5’ end transcript
- Overlaps with or is near the promoter/transcription start site

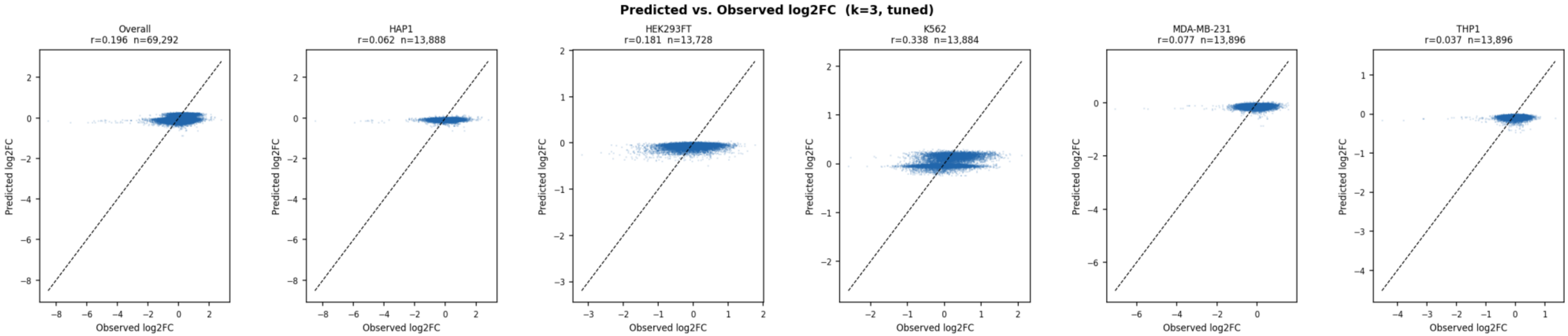
3’ end/polyadenylation signal

- 3’ UTR region
- Contains stability elements and polyadenylation signals

Model eval — XGBoost baseline k=3

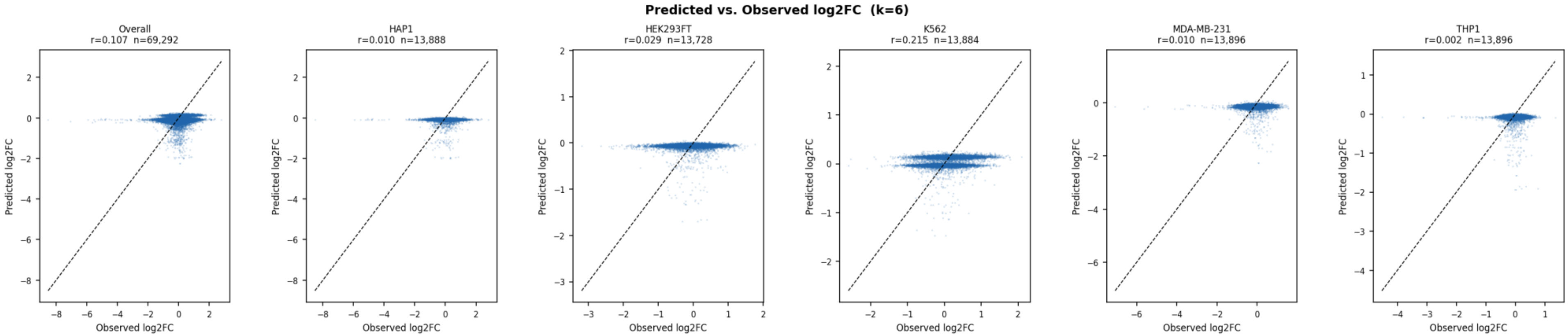


Model eval — Optuna hyperparameter tuning k=3

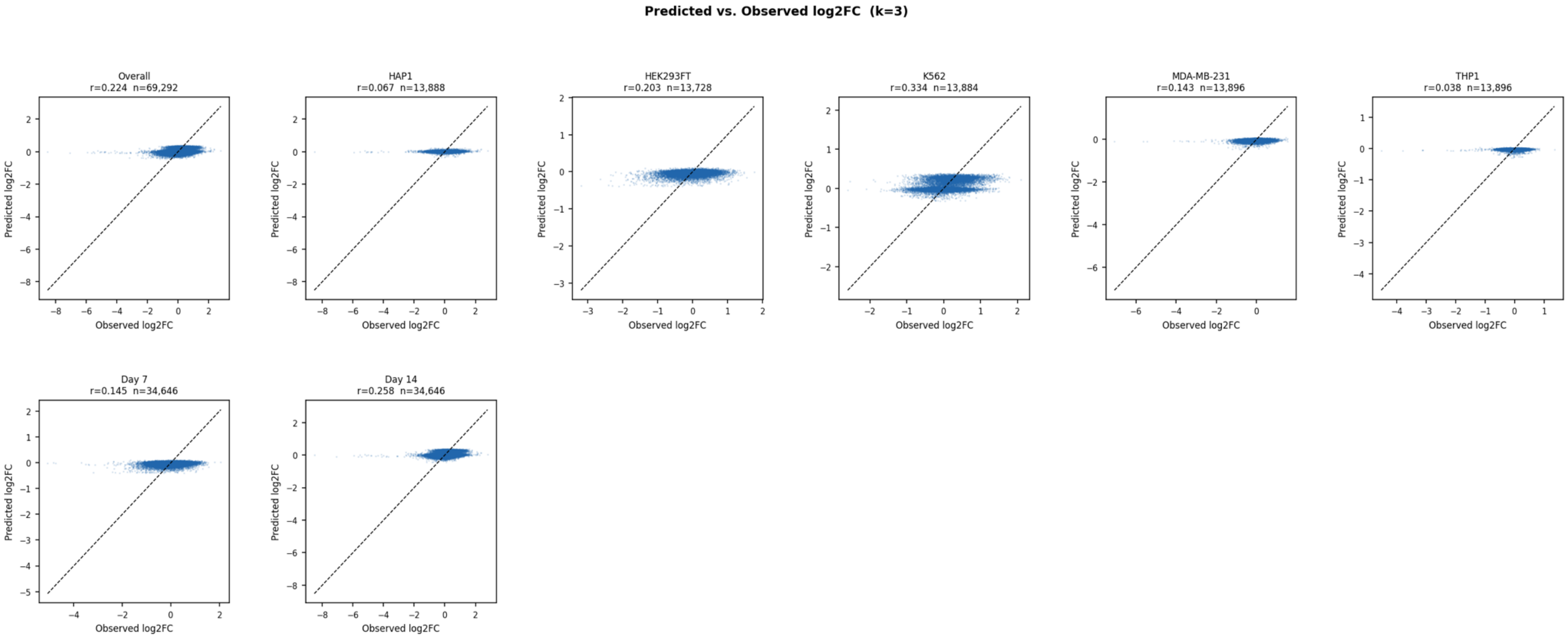


Split	Pearson r	Spearman ρ	R^2
Overall	**0.196** (+6.6%)	0.141 (–)	**0.019** (+105%)
K562	0.338	0.312	0.093
THP1	0.037	0.020	−0.066
day 7	0.111	0.077	0.002
day 14	0.237	0.196	0.026

Model eval — Vanilla k=6

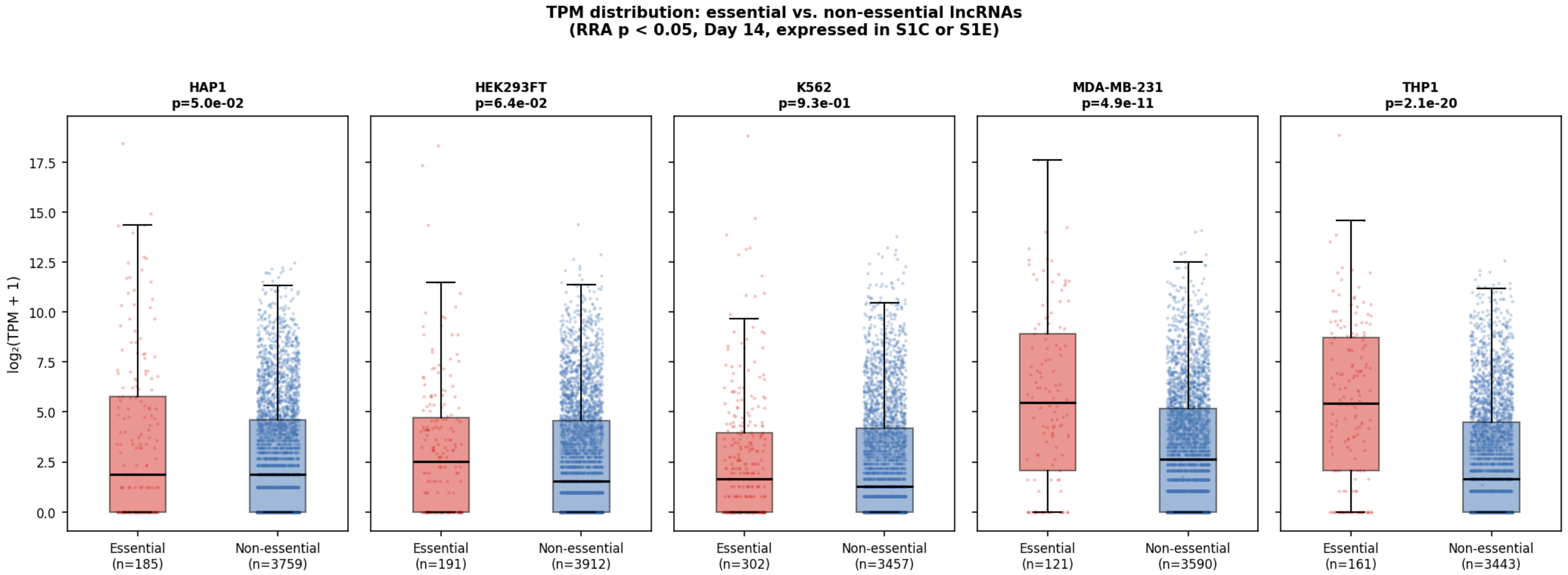


Transcript overlap



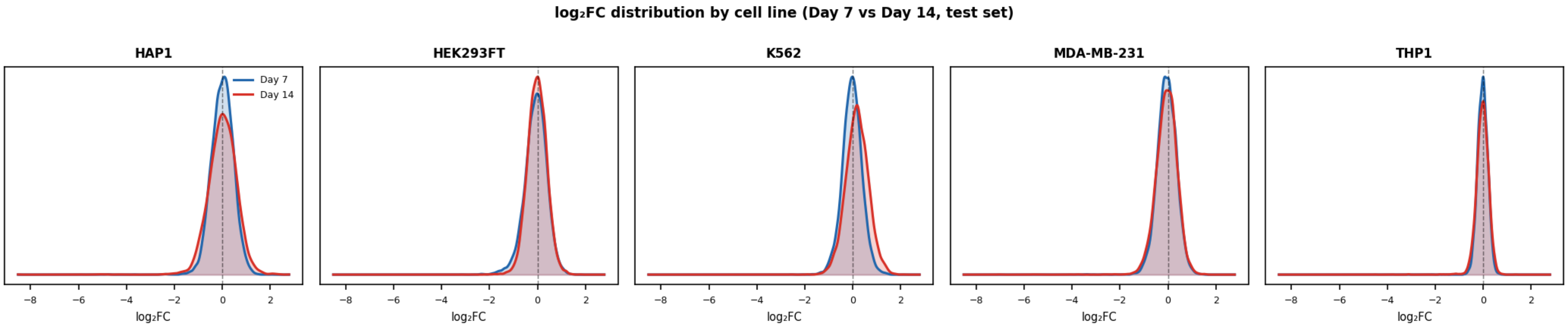
Essential lncRNAs have higher TPM – but expression alone is insufficient

Essential lncRNAs have statistically higher TPM (Wilcoxon rank-sum $p < 0.05$ in 4/5 cell lines) but Spearman $\rho = -0.017$ to -0.089, explaining <1% of variance — ruling TPM out as a standalone predictor.



Sanity check

Distribution of observed log2FC values



Full model benchmark table – Genomic overlap

All models evaluated on chr1 hold-out (n = 69,292) · pooled across all cell lines + days

Model	Pearson r	Spearman ρ	RMSE	R^2
Null (mean)	—	—	0.436	−0.022
Ridge α=100	0.094	0.079	0.431	0.002
Lasso α=0.01	0.105	0.091	0.434	−0.012
XGBoost k=3 Optuna ★	**0.226**	**0.172**	0.420	0.050

★ Best pooled model — all per-cell-line × day models use k=3 Optuna-tuned hyperparameters

Full model benchmark table – Transcript overlap

All models evaluated on chr1 hold-out (n = 69,292) · pooled across all cell lines + days

Model	Pearson r	Spearman ρ	RMSE	R^2
Null (mean)	—	—	0.436	−0.022
Ridge α=100	0.089	0.079	0.431	-0.001
Lasso α=0.01	0.105	0.091	0.434	−0.012
XGBoost k=3 Optuna ★	**0.225**	**0.173**	0.420	0.050

★ *Best pooled model — all per-cell-line × day models use k=3 Optuna-tuned hyperparameters*

The signal is real — but only non-linear models capture it

0.172

XGBoost Spearman ρ

vs 0.079 Ridge (~2×)

≈ 0

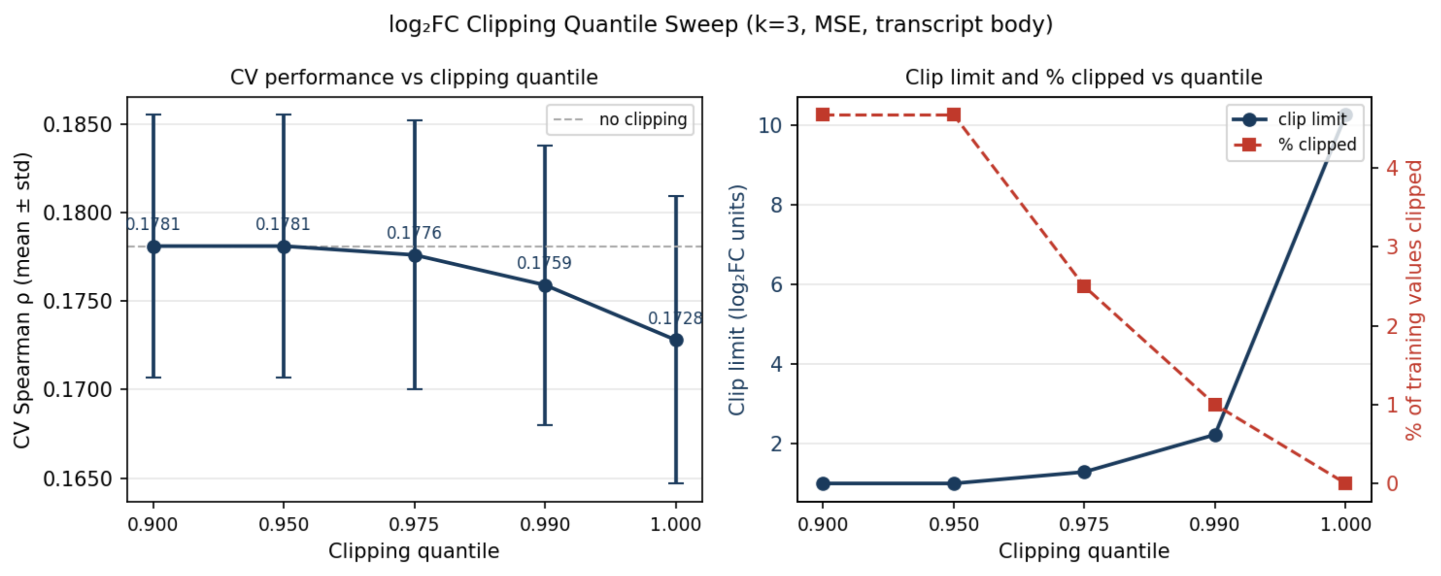
Linear models R^2

barely beat null predictor

Genomic ≈ transcript body features — both ~2× linear models. The bottleneck is model class, not feature set.

log2FC clipping sweep

Symmetric quantile clipping of training targets · LOCO-CV Spearman ρ



✓ Clipping consistently improves CV ρ (+0.005 at best)

⚡ Plateaus at q=0.95 — both 0.95 and 0.90 hit the minimum_limit=1.0 floor, clipping exactly 4.67% of data

★ Recommended default: --clip-quantile 0.95 for all future runs

Liang et al., data processing pipeline

Design library

~56,000 guides vs
~5,500 lncRNAs

Transduce

Lentivirus → 5 lines,
low MOI

Induce & sample

Dox-induced;
sample D0 / 7 / 14

Depletion

Essential guides
deplete

Score (RRA)

RRA → gene-level
log2FC

Open Questions

1. Why do HAP1 and THP1 stay near-random even with per-cell-line training?
2. Why does HEK293FT's day-effect flip opposite to K562?
3. Does ChatNT fine-tuning beat XGBoost?
4. Guide-level vs lncRNA-level prediction? What sequence features actually drive essentiality?

What's been built (jeonglab-bcm/lncFit)

💾 Data pipeline

JSONL + gzip (Git LFS)
Chromosome train/test splits
ScreenRecord schema + versioning

🔢 Feature engineering

k-mer freq vectors (k=3,6)
Signed overlap encoding
Genomic + spliced transcript seqs

🤖 Model training

XGBoost + LOCO-CV
Optuna TPE hyperparameter tuning
Per-cell-line × day specialization

📊 Evaluation

Null / Ridge / Lasso baselines
Per-cell-line + per-day metrics
Scatter plots + CSV bundles

🛠 Code quality

Modular package (lncfit/)
Shared cv.py, metrics.py, plotting.py
Memory profiler + provenance JSON

🔬 Key fixes

k=6 OOM fix (float32 prealloc)
revcomp bug in signed overlap
Corrected Liang et al. gRNA library